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Assignment 4 Comp-4540

Prove that 3-SAT $\propto$ TT

TT is defined in the question.

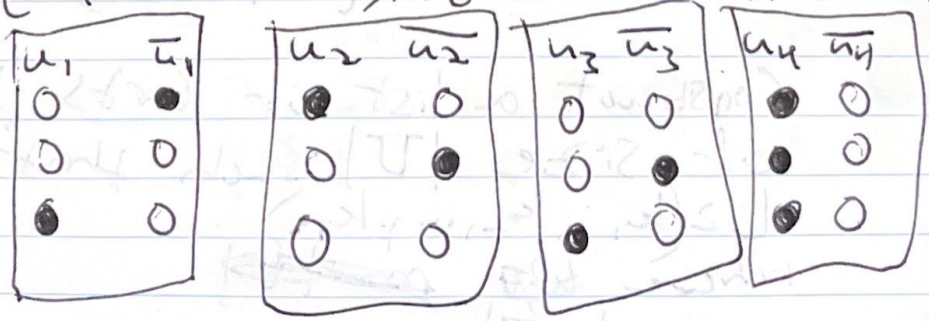
Idea: To transform the input from 3-SAT to TT each card in TT represents a literal in 3-SAT and every row of holes represents a clause in 3-SAT. Column 1 of a card represents the literal for each clause and column 2 represents the literal negated in the clause. ~~If punched out that literal is present in the clause~~ If a hole is filled in for that given clause and for the literal, that literal is present in that clause.

Example: Using literals: $u_1, u_2, u_3, u_4 \in V$

$|V| = 4 = n$ Let there be 3 clauses in the Problem instance.

$(\neg u_1 \vee u_2 \vee u_4) \wedge (\neg u_2 \vee \neg u_3 \vee u_4) \wedge (u_1 \vee u_3 \vee u_4)$

$n = 3$



Because The True card is on top, one side is already completely filled. Additionally, the two faces of the cards are indistinguishable so, if either side is covered it is a yes-instance for that clause. 3-SAT having atleast one True value in each clause corresponds to each row of cards having atleast One hole filled for that row.

Actual Proof:

Find a polynomial reduction from 3-SAT to IT.

Let $C = c_1 \wedge c_2 \wedge \dots \wedge c_m$ be a 3-CNF logical expression such that $c_j = (u_{j,i} \vee u_{j,i} \vee u_{j,i})$, $1 \le j \le m$, and $\exists i : 1 \le i \le n$, $n = |U|$,

$u_{j,i} \in U \cup \overline{U}$, $1 \le i \le n$, where U is the set of variables in C and $\overline{U} = \{\bar{u} \mid u \in U\}$.

Construct a list of cards, L, of size $|U|$ such that $L = (k_1, k_2, \dots, k_n)$ where ~~$k_i \in \{$~~ ~~$\}$~~ $n = |U|$.

For Each card k_j, $1 \le j \le m$,
has 2 columns of holes
of size m.
For each hole and for
each clause C_j, $1 \le j \le m$,
if u_{ji} is present in C_j then
fill the k_ith cards j^{th} hole
on the left
else if $\overline{u_{ji}}$ is present in C_j
then fill the k_ith cards j^{th}
hole on the right

Now we have a list of
cards corresponding to C.

Time complexity:

Constructing L takes $|V| = n$
entries and each card comparing
twice for each clause, so $2m = 2|C|$
times. (worst case).

$= O(|V| \times 2|C|) = O(|V| \times 2n)$
$= O(n)$ since $|V| \le 3n$
from being 3-CNF.
$\therefore$ Construction is in polynomial time.

C is satisfiable iff L is stalled in a way where you cannot see through holes.

⇒) Suppose C is satisfiable and t be a truth assignment satisfying C.

if $t(u_{ji}) = \text{true}$, $u \in U \cup \bar{U}$, ~~then as that is a part of a clause (joined by V) then the whole clause is true. So, then~~ the u_{ji}^{th} hole is covered on the left or right when u_{ji} is present in C.

if $t(u_{ji}) = \text{false}$, then the u_{ji}^{th} hole is not covered regardless if it is present in C or not.

Since C is satisfiable, for each C_j at least one $t(u_{ji}) = \text{true}$ in that clause.
⇒ For every j, $1 \le j \le m$, $\exists i$, $1 \le i \le n$ $u_{ji} = \text{true}$ under t.
i.e. For every j, $1 \le j \le m$, $\exists i$, $1 \le i \le n$ hole u_{ji} is covered in the left or right column.

~~⇒ every card has atleast one hole covered~~

⇒ Every row of holes (clauses) has atleast one card (variable) which covers the hole.
Since, the truth card covers one side already and you are free to flip any of the cards, the holes can be fully covered.

⇐ Suppose L is fully covered.
Then, there exists atleast one card that covers a hole for each hole.

Define $t: U \cup \bar{U} \to \{true, false\}$
such that
$t(u) = \begin{cases} true \text{ if left hole filled} \\ false \text{ if left hole empty} \end{cases}$

Since each hole is covered by a card ~~$c_j, c \in U \cup \bar{U}, 1 \le j \le m, 1 \le i \le n$~~

~~Every~~ hole c_j has been covered by card k_i such that
$t(c_{ji}) = true$ for some i.
$\therefore \forall j, 1 \le j \le m, C_j = true$ under t.
$\therefore C$ is true under t. C is satisfiable. □